

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - US-based, F-1 OPT (open to sponsorship); willing to relocate to Chicago

PROFESSIONAL SUMMARY

Full-stack engineer who ships production AI features end-to-end. Built and published a multi-LLM Chrome extension (Manga Lens, Chrome Web Store) integrating four AI vision providers; React/TypeScript + FastAPI full-stack apps with structured prompt orchestration; agentic LLM workflows with schema-validated JSON contracts. Applied-LLM track record from healthcare: Retrieval-Augmented Generation (~35% precision gain), agentic workflows (>30% hallucination reduction), embeddings/semantic retrieval, evaluation pipelines, and Docker-based deployment. Master's in Computer Science (Kent State, 2025). Comfortable owning frontend, LLM agents, databases, and event-driven services with bias toward shipping in hours, not days.

CORE COMPETENCIES

TypeScript • React • Node.js • Full-stack Web • LLM Agents • RAG & Semantic Retrieval • Embeddings • Python / FastAPI • Databases (PostgreSQL, T-SQL, SQLite) • Event-driven Services • Docker • AI-native development tools (Claude, Cursor)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting

- Shipped **full-stack AI features** end-to-end: data ingestion → LLM workflow → FastAPI service → Docker rollout; reduced post-deployment defects ~**30%** via structured logging, load simulation, and evaluation audit trails.
- Built **Retrieval-Augmented Generation** with recursive semantic chunking and transformer embeddings; improved retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**.
- Developed **agentic LLM workflows** with structured outputs, tool discipline, and grounding rules; improved response stability ~**25%**; benchmarks and eval harnesses gated each release.
- Owned **predictive ML pipelines** (scikit-learn, XGBoost); improved recall **15-20%** on high-risk categories while holding precision >**90%**.
- Used **Claude and AI-native dev tools daily** to compress build cycles - drafting, refactoring, eval scaffolding, and documentation pipelines.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Video Analytics

- Built a **transformer-based summarization system** across 5,000+ sessions; hierarchical long-context summarization cut manual review time **60-70%** (hours → <5 min/video).
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinations ~**30%**, raised answer accuracy >**85%**.
- Deployed as a **Flask API** with a lightweight web interface for non-technical staff.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern)

- Built **CI/CD pipelines with Jenkins** for version-controlled SQL and stored-procedure deploys; reduced deployment errors >**30%** and release cycle time ~**35-40%**.
- Tuned T-SQL stored procedures and partition-aware indexing; cut query execution time ~**20%** and retrieval latency ~**25%**.
- Built performance dashboards (DMVs + Grafana) for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, **captureVisibleTab** capture pipeline with viewport-slice screenshots and coordinate remapping for tall webtoon panels. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind a common interface, per-provider payload handling (WebP for cloud, JPEG for Ollama), 7-day translation cache, per-domain selector configs for 29 sites, and a privacy policy.

TypeScript, Manifest V3, service workers, multi-LLM orchestration, real-time vision pipelines

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate image models). Introduced intermediate **structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success **~25-30%** over naïve direct-prompt orchestration.

React, TypeScript, Vite, FastAPI, structured prompt orchestration

Agentic Healthcare Claims & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations; RAG-grounded validation against vector-indexed policy docs, ANN duplicate detection, and explainable audit-ready reasoning traces.

Multi-agent LLM, schema contracts, RAG, vector search, evaluation

Geospatial Road Network Analysis & Community Detection

Parsed OpenStreetMap XML into a weighted **NetworkX** DiGraph; Dijkstra shortest-path routing, betweenness centrality on a 20k-node subgraph, Louvain community detection, and MST extraction. **~30% runtime reduction** via subgraph scoping - algorithmic rigor on real graph data.

Python, NetworkX, graph algorithms, geospatial

Patient Records Management Web App

Bottle + SQLite relational CRUD app with normalized **patients/diagnoses** schema, foreign-key constraints, application-layer cascade deletes, and **LEFT JOIN + GROUP_CONCAT** aggregation for single-query patient+diagnosis retrieval.

Python, Bottle, SQLite, relational design

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: TypeScript, React, Node.js (adjacent), Python, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), PHP

LLM & GenAI: LLM application development, agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, grounding, embeddings, vector/semantic search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging, Grafana, event-driven service design

AI-native dev tools: Claude, Cursor, multi-LLM orchestration